RWANDA ESTABLISHMENT DISTRIBUTION AND EMPLOYMENT ANALYSIS SYSTEM

INSY 8311 – Database Development with PL/SQL | Phase 0 Project Proposal

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GitHub repository: [https://github.com/Mouminat/Rwanda-establishment-employment-analysis.git]	Domain: Business and Entrepreneurship

1. Problem Statement

Information about the number, location, economic activities and employment characteristics of establishments is useful for understanding Rwanda's business environment. However, decision-makers, researchers and students may need to consult different statistical tables to compare these characteristics across districts and activity groups. The National Institute of Statistics of Rwanda (NISR) Establishment Census 2023 provides establishment-level information, but this project will organize selected variables in a structured relational database and make common comparisons easier to retrieve and review. The NISR report records 269,326 establishments and 927,739 workers nationally, while the public microdata catalogue lists 269,194 cases and 68 variables. These are different published counts and will be reported separately rather than treated as identical.

2. Proposed Solution

The project proposes an Oracle relational database system that stores and organizes available anonymized establishment census data. Users will be able to retrieve and summarize establishments by district, province, major economic activity, establishment size and selected employment characteristics. The system will produce queries and summary reports to support comparisons across geographic areas and business activity groups. This is an analytical database and reporting solution; it is not a business registration, licensing, inventory or transaction-processing system.

3. Data Source and Research Basis

The primary dataset is the Rwanda Establishment Census 2023 microdata published by NISR: https://microdatabk.statistics.gov.rw/catalog/112. Its catalogue lists 269,194 cases and 68 variables, including geographic location, economic activity, establishment size, formal/informal status and employment variables. Access to the microdata may require a free account on the NISR microdata catalogue. The dataset will be downloaded and inspected before the database design is finalized. A suitable subset containing at least several hundred records will be selected if required by the project scope, and the selection method and exclusions will be documented. Only anonymized variables needed for the project will be used.

4. Objectives

- Design a relational database from the selected census variables and normalize the design to Third Normal Form (3NF).
- Load and validate an adequately sized set of anonymized establishment records.
- Develop SQL and PL/SQL operations for data retrieval, summaries, validation and controlled updates where appropriate.
- Generate district-, activity-, size- and employment-based reports.
- Present key summary results through an interactive analytical dashboard connected to the project database.

5. Planned Database Design

The initial logical design may include ESTABLISHMENT, DISTRICT, ECONOMIC_ACTIVITY and ESTABLISHMENT_SIZE entities, with employment measures associated with establishment records or modeled separately if the data structure justifies it. These are provisional entities, not a final schema. Primary keys, foreign keys, attributes and relationships will be confirmed only after examining the downloaded dataset, its codebook and available identifiers. The final design will avoid unnecessary repetition and will be normalized to 3NF.

6. Planned PL/SQL Features

The project plans to use stored procedures for controlled data loading and maintenance; functions for calculating establishment and employment summaries; cursors for processing and presenting grouped records; exception handling for invalid or missing data; and triggers with an audit table to record selected database changes. Bulk processing may be used where it provides a clear benefit during data loading or batch operations. The exact implementation will be adjusted to the final schema and course requirements.

7. Planned Innovation and Expected Results

The planned innovation is an interactive analytical dashboard connected to the Oracle database. It will allow users to view and filter summarized establishment and employment information by district, activity and establishment size. The dashboard will complement, not replace, the database; core data rules and processing will remain in Oracle PL/SQL. Expected results include a normalized database, documented PL/SQL components, reproducible summary reports and a working dashboard that makes selected census indicators easier to explore. A limitation is that the 2023 census is a snapshot and should not be treated as real-time business information.

8. References (APA 7th edition – verify each source before submission)

National Institute of Statistics of Rwanda. (2025). Establishment Census 2023. https://www.statistics.gov.rw/data-sources/censuses/Establishment-Census/establishment-census-2023

National Institute of Statistics of Rwanda. (2024). Rwanda Establishment Census 2023: Microdata catalogue. https://microdatabk.statistics.gov.rw/catalog/112

National Institute of Statistics of Rwanda. (n.d.). Business and establishment statistics. https://alpha.statistics.gov.rw/statistical-publications/business-establishment-finance-trade/business-establishment-finance

Oracle. (n.d.). Oracle Database documentation: PL/SQL language reference. https://docs.oracle.com/en/database/oracle/oracle-database/

[Add two research papers you have personally read and verified. Do not submit this proposal with this placeholder remaining.]